

18-button joystick mouse

WarMouse's 18-button joystick mouse, rechristened the Meta, will now ship with a high-resolution laser sensor instead of an optical sensor. The Meta is one of the first computer mice to incorporate an analogue joystick and is the first mouse to permit the use of the joystick as a digital keyboard interface. It has been alternately described as "the most insane mouse ever," "a Photoshop gag" and "too much of a good thing."

In the three digital joystick modes,



WarMouse from OpenOffice

the user can assign up to 16 different keys or macros to the joystick. In analogue joystick mode, the player can select between having up to seven joystick buttons available or using all 18 buttons as keyboard and mouse commands instead.

The WarMouse Meta also includes 18 programmable mouse buttons with double-click functionality; high-resolution laser sensor with adjustable resolution from 100 to 5,600 DPI/CPI; five assignable button modes: Key, Keypress, Macro, Mouse and Mouse-Key Combo; analogue Xbox 360-style joystick with five analogue and digital modes; clickable scroll wheel; 512k of flash memory; 63 on-mouse application modes with hardware, software and autoswitching capability; and 1024-character macro support.

The Meta is compatible with Windows, Linux and Macintosh operating systems. It is expected to hit the market in the first quarter of 2010.

BlindAid system

The Massachusetts Institute of Technology (MIT) has developed a new device that could give the visually impaired

the same kind of benefit that sighted people get from visual mapping applications such as Google maps. The BlindAid system allows blind people to "feel" their way around a virtual model of a room or building, familiarising themselves with it before going there.

The BlindAid stylus functions much like a blind person's cane, allowing the user to feel virtual floors, walls, doors and other objects. The stylus is connected to a computer programmed with a three-dimensional map of the room. Whenever a virtual obstacle is encountered, the computer directs the stylus to produce a force against the user's hand, mimicking the reaction force from a real obstacle.

To successfully use such a system, the visually impaired person must have a well-developed sense of space. For those people, this offers the prospect of being able to preview an environment and have a sense of what's in the environment, the shape of it and where landmarks are located.

Hot electrons could double solar power

Researchers at Boston College have experimentally verified a theory suggesting that hot electrons could double the output of solar cells. They observed the 'hot electron' effect in a solar cell for the first time and harvested the elusive charges using ultra-thin solar cells – thereby opening a potential avenue to improved solar power efficiency.

When light is captured in solar cells, it generates free electrons in a range of energy states. But in order to snare these charges, the electrons must reach the bottom of the conduction band. The problem has been that these highly energised 'hot' electrons lose much of their energy to heat along the way.

By using ultra-thin solar cells – a film fewer than 30 nanometres thick – the team developed a mechanism that can extract hot electrons in the moments before they cool – effectively opening a new 'escape hatch' through which they typically don't travel. The

team's success centered on minimising the environment within which the electrons are able to escape.

The ultra-thin cells demonstrated overall power conversion efficiency of approximately 3 per cent, using absorbers one fiftieth as thick as conventional cells.

Nanosensors could detect cancer!

A team of researchers at Yale University has used nanosensors to measure cancer biomarkers in whole blood for the first time. The new findings could simplify the way physicians test for biomarkers of cancer and other diseases.

Researchers used nanowire sensors to detect and measure concentrations of two specific biomarkers: one for prostate cancer and the other for breast cancer.

Nanosensors have been around for the past decade, but they only worked in controlled, laboratory settings. This is the first time they have been used with whole blood, which is a complicated solution containing proteins and ions and other things that affect detection.

To overcome the challenge of whole blood detection, the researchers developed a novel device that acts as a filter, catching the biomarkers – in this case, antigens specific to prostate and breast cancer – on a chip while washing away the rest of the blood. Creating a buildup of the antigens on the chip allows for detection down to extremely small concentrations, of the order of picograms per millilitre, to within an accuracy of plus or minus 10 per cent. This is the equivalent to being able to detect the concentration of a single grain of salt dissolved in a large swimming pool.

The new device could also be used to test for a wide range of biomarkers at the same time, from ovarian cancer to cardiovascular disease.

Dual-screen e-book reader

Here comes a dual-book called the enTourage eDGe, which combines the functions of an e-reader, netbook,